Package Assignment and Routing System (PARS)

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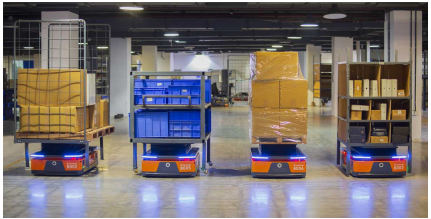
Introduction

- Suppose you're a warehouse operator serving a city.
- Your job is to distribute goods from the warehouse using delivery trucks to transport them to their final destination in an efficient manner.
- However, a single truck can carry far more than just one delivery at a time.
- As such, we must design efficient routes for these trucks.

Motivation

For our project we'll be using the analogy of delivering packages to homes.

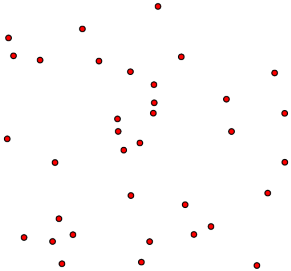
- Consider, school bus routing, human and robotic warehouse pickers/stockers, waste collection, snow plowing, coordination of emergency services, etc.
- Finding an optimal solution is computationally expensive. Must resort to heuristics to find a reasonable solution in a reasonable amount of time.
- AI revolves around this idea of making good enough decisions in a reasonable amount of time.



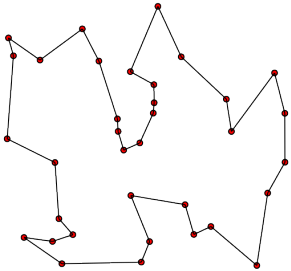
Foundational problem

Traveling salesman problem (TSP) — "Given some origin city, what's the shortest possible route that visits each city exactly once and returns to the origin city?"

- Modeling this as a graph theory problem, we can assume a complete graph K_n where nodes are cities and edge weights are pre-calculated distances.



Unsolved



Solved

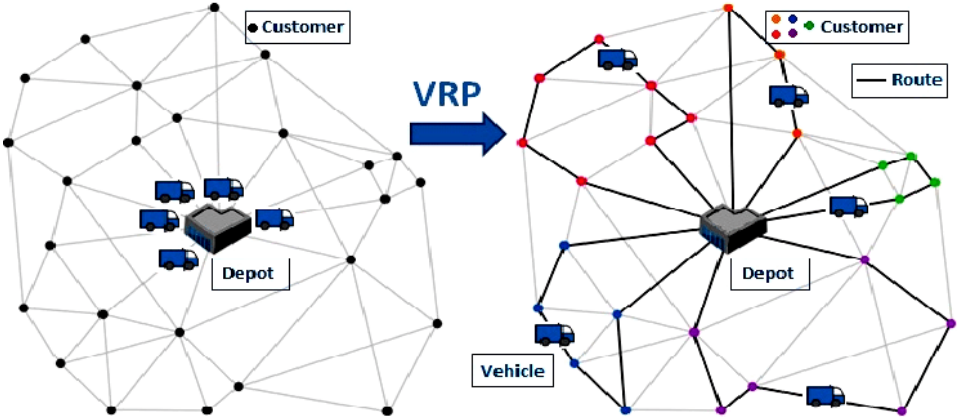
Vehicle routing problem

Vehicle routing problem (VRP) — Generalization of the traveling salesman problem which asks "what is the optimal set of routes for a fleet of vehicles to traverse in order to deliver to a given set of customers?"

- Typically, goal is to minimize the total distance traveled by all vehicles, however, many more metrics might be considered.
- Can include any number of constraints that must be satisfied—a vehicle can only carry so much weight, a driver can only work for so long.
- Can be thought of as attempting to solve multiple traveling salesman problems or a multi-agent search problem.

VRPs usually still assume a complete graph where actual road routing is abstracted away. However, we will perform our VRP over graph which models a real-world road network.

VRP overview



Gupta, Ashima & Saini, Sanjay. (2017). An Enhanced Ant Colony Optimization Algorithm for Vehicle Routing Problem with Time Windows. 267-274. 10.1109/ICoAC.2017.8441175.

Formal definition

Let $G = (V, E)$ be a connected, weighted graph where

- vertices represent intersecting roads and/or customers,
- vertices also possess latitude/longitude information (x, y),
- all trucks start and end at a central warehouse described by a node $w \in V$,
- $C \subseteq V \setminus \{w\}$ is the set of destinations (customers) the trucks must visit,
- edges represent the existence of a road connecting two vertices where edge weights describe the distance between the two nodes.

Let T be a set of trucks each with fixed capacity k where the total customers a single truck visits cannot exceed k.

We define function $\text{PARS}(G, w, C, T, k)$ which outputs a mapping $T \to \{(w, \ldots, w) \subseteq V, \ldots\}$ which describes the route of each truck starting at w and ending at w. The goal of this function is to minimize the total distance traveled, the sum of edge weights, across all trucks.

Our proposal

1. Solve a single TSP (this is equivalent to having a single truck with an infinite capacity) using nearest-neighbor heuristic and A* search.
2. Effectively, partition this route among vehicles
3. Use k-means clustering to partition nearby customers and solving a TSP for each cluster.
4. Use modern AI techniques like reinforcement learning to attempt to find more optimal solutions over our classical approaches.

Our primary goal is to reduce the distance traveled by all trucks in total. We will record this metric, then compare and contrast different techniques and iterations of our solution.

Main reference

Progress so far

An 8-customer TSP solved using nearest neighbor heuristic (for picking the next node to visit) and A* search (for navigation between nodes) in Albany.

Thank you!